# Aliağa Demo Steel Works (fictional)

Installation Mass & Energy Balance Boundary Diagram

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| DIAGRAM ID: PFD-2026 | DATE: 2026-01-15 |
| ROUTE: Scrap EAF -> CCM -> Reheat -> Rolling | PLANT ID: TR-ALIAGA-DEMO-001 |

Stage 1: Scrap Yard & Basket Preparation (Sorting, Radiometric Screening, Charging)

Stage 2: Electric Arc Furnace (100 MVA AC Furnace, Oxy-Fuel Burners, Carbon Injection)

Stage 3: Ladle Refining Furnace (Secondary Metallurgy, Desulphurisation, Alloying)

Stage 4: 6-Strand Continuous Casting Machine (Billet Production, Cut to Length)

Stage 5: Billet Reheating Furnace (Walking Beam, Low-NOx Burners, Natural Gas Fired)

Stage 6: Hot Continuous Rolling Mill (Roughing, Intermediate, Finishing Mill for Rebar)

System boundaries align strictly with European Commission CBAM Implementing Regulation (EU) 2025/2547 rules for scrap-based electric arc furnace production of iron and steel products (CN 72142000).